

MANUELA, Antarctica

WMO: 898640

Lat: 74.946S

Lon: 163.687E

Elev: 262

StdP: 14.56

Time Zone: 12.00 (NZL)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-35.6	-32.9	N/A	N/A	N/A	N/A	N/A	N/A	96.6	-5.9	90.4	-12.1	16.1	290	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	36.6	N/A	34.5	N/A	32.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 76.6	61.8	51.3	DB	-41.0	42.6	5.2	3.1	-44.7	44.8	-47.8	46.6	-50.7	48.4	-54.5	50.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	25.6	12.5	-3.8	-12.3	-13.7	-13.2	-16.3	-16.5	-12.6	-3.4	14.0	25.5		
	DBStd	17.37	4.27	6.61	7.61	8.10	8.83	8.70	9.80	9.82	8.77	6.93	6.97	4.25		
	HDD50	18698	756	1049	1667	1868	1975	1896	2054	2062	1879	1655	1079	758		
	HDD65	24173	1221	1469	2132	2318	2440	2346	2519	2527	2329	2120	1529	1223		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	17.3	11.3	16.2	26.7	21.0	20.3	21.7	19.5	17.0	13.4	19.2	12.5	8.2		
	PrecAvg	9.9	1.1	0.9	0.9	0.7	0.8	0.9	0.7	0.9	0.7	0.7	0.7	0.8		
(15) Precipitation	PrecMax	13.1	2.3	2.6	2.1	1.6	1.9	1.7	1.4	2.2	1.9	1.6	2.3	1.7		
	PrecMin	5.3	0.3	0.4	0.4	0.1	0.1	0.2	0.1	0.1	0.0	0.1	0.2	0.1		
	PrecStd	1.9	0.5	0.4	0.4	0.4	0.4	0.4	0.3	0.6	0.5	0.4	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	30.3	18.9	16.1	13.2	13.7	20.9	17.1	13.3	18.0	32.3	39.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	36.8	26.9	13.8	8.8	6.8	9.0	8.4	8.8	9.3	12.6	27.9	37.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	34.6	24.0	10.0	3.5	2.6	3.7	1.5	2.0	4.7	9.3	25.8	35.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	32.7	21.6	6.6	-0.9	-1.2	-1.0	-3.0	-2.3	0.3	6.3	23.7	33.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	MDBR	11.9	9.9	8.8	8.8	9.1	9.7	9.4	10.2	10.0	11.1	12.1	12.2		
		MCDBR	14.0	12.5	11.0	13.3	11.6	13.2	13.8	13.7	12.9	13.4	13.5	14.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	Clear-Sky Solar Irradiance	taub	0.240	0.225	0.184	0.111	N/A	N/A	N/A	0.109	0.173	0.199	0.221	0.237		
(40) All-Sky Solar Radiation	taud	2.303	2.299	2.242	1.904	N/A	N/A	N/A	N/A	1.882	2.191	2.277	2.284	2.293		
		Ebn at Noon	312	292	261	133	0	0	0	126	261	303	318	319		
	Edh at Noon	31	25	17	4	0	0	0	0	4	17	26	31	33		
		RadAvg	2667	1613	659	96	0	0	0	25	399	1300	2467	3099		
(44) All-Sky Solar Radiation	RadStd	59	63	30	5	0	0	0	0	2	22	39	67	78		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page